

College Project Management & Review Portal (With Machine Learning)

Project Overview

This project is a web-based academic platform designed to manage the complete lifecycle of college projects. Students submit projects, tag preferred guides, receive reviews, and track progress digitally. Machine Learning is integrated to add intelligence, prediction, and automation to the system.

Problem Statement

Traditional project handling in colleges is mostly manual and fragmented. Students submit projects through emails or physical copies, guides lack real-time progress visibility, and repeated project topics often occur. This results in delays, miscommunication, and inefficient evaluation.

Proposed Solution

A centralized project management portal where students submit projects and tag their preferred guides. Guides approve or reject requests and provide structured reviews and evaluations. Admins monitor overall progress and system analytics.

User Roles

Student: Submit project details, tag guide, upload reports, track review status.

Guide: Approve or reject project requests, review submissions, provide feedback, assign marks.

Admin: Monitor users, deadlines, project status, and analytics.

Key Feature: Student Tags the Guide

During project submission, the student selects a preferred guide from the available list. The selected guide must approve the request before the review process begins. This ensures proper consent, workload balance, and a realistic academic workflow.

Machine Learning Features

- Project Delay Prediction using Random Forest or Logistic Regression
- Duplicate Project Topic Detection using NLP (TF-IDF and Cosine Similarity)
- Student Performance Risk Analysis based on academic and project activity data

System Architecture

Frontend (React) → Backend API (Node.js / Flask) → Database (MongoDB) → ML Models

Technology Stack

Frontend: React.js

Backend: Node.js / Flask

Database: MongoDB

Machine Learning: Scikit-learn, NLP techniques

Authentication: JWT

Conclusion

This project provides a structured, scalable, and intelligent solution for managing academic

projects. By combining full-stack development with machine learning, the system improves transparency, monitoring, and decision-making in college project evaluation.